

A Low-complexity, FPGA-based Maximum Power Point Tracking Circuit, for Wide Range Light-energy Harvesting

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Abstract—This work presents the design of a low-complexity FPGA-based maximum power point tracking (MPPT) circuit for wide power range light-energy harvesting applications. The circuit relies on a self-oscillating dc-dc boost (SOB) converter for the power conversion, which is controlled via an analog voltage. A custom-made digital-to-analog (D/A) converter circuit is used to produce the required analog voltage, while a resistor-free current-to-frequency (C/F) converter circuit provides feedback of the SOB converter's output power. The FPGA-based digital part of the circuit controls the D/A converter and implements the perturb and observe MPPT logic using the frequency feedback of the C/F converter. The circuit was constructed and measured using discrete components. The digital MPPT control has been described in Verilog HDL and implemented on an Intel Cyclone 10CL016 FPGA. The circuit works in a very wide input power range from 15 μ W up to 1 W with constant efficiency and is suitable for light energy harvesting both at indoor and outdoor lighting conditions. Experimental results prove this implementation to be suitable for on chip integration.

Keywords—current to frequency converter; self-oscillating dc-dc boost converter; digital to analog converter; digital signal processing; energy harvesting; FPGA; indoor-outdoor light; low-power; MPPT; wide power range

I. INTRODUCTION

Solar energy is the most widely used energy source, supporting a large number of autonomous devices. Wireless sensor nodes and portable or wearable devices are such examples [1]. As the power consumption of modern electronic circuits decreases, the uninterrupted operation of an autonomous device can be supported by harnessing energy from low energy sources such as indoor light [2]. Unfortunately, harvesting circuits optimized to work at the lowest power levels, usually offer limited working power ranges [3]. In realistic conditions, an autonomous device designed for indoor lighting conditions may be exposed to outdoor light as well [4]. In this scenario, it could use the extra energy to increase either its autonomy (life-expectancy), or its performance. The power output of a photovoltaic (PV) panel working both at indoor and outdoor lighting conditions can vary in

a very wide power range, thus, a harvesting circuit that is able to work in an equally wide range is required [5]. Other important parameters regarding the harvesting circuit are the power conversion efficiency as well as its power consumption [6].

To further increase the conversion efficiency the most efficient harvesting circuits perform input matching operations in order to track the maximum power point (MPPT), and various methods have been proposed for achieving it [7][8]. The most efficient ones require feedback of the working power in a closed-loop circuit, in order to adjust their operation [9]. Resistor-based power sensing is not suitable for wide range targeted designs, as the voltage drop on the sensing resistor increases with the input power, leading to huge power losses and low power conversion efficiency at high input power levels [10]. Alternative methods have been proposed which rely on time-based sensors [11][12]. The time-based power sensing works without interfering with the range, or the efficiency of the harvesting circuit, and additionally, it minimizes the required components and power consumption of the MPPT control circuit [13].

Fully analog MPPT control designs have been proposed, which offer good MPPT accuracy and very low power consumption [14]. However, most modern MPPT solutions embed a digital control circuit, as more sophisticated MPPT schemes can be adopted this way, with higher accuracy and similar power consumption [15]. Many published works present FPGA- or microcontroller-based designs, which are proven to be powerful tools for the development and verification stages of the method or the design [16]. Though, ultra-low power MCUs are commercially available, the on-chip integration is the optimal solution (application-specific integrated circuit – ASIC), minimizing the size of the final system as well as its power consumption [17].

The most important part of an energy harvesting system is the power conversion unit, and various different topologies are available (boost converter, buck converter etc.). The majority, require a digital signal for the control and usually pulses with fixed frequency and varying width

(pulse width modulation – PWM), or pulses with varying frequency (pulse frequency modulation – PFM), are used [18–20]. The demand for size minimization leads to the use of small inductances and high working frequencies, that drastically increase the power consumption of the digital circuit. Depending on the design specifications, a typical clock frequency for a digital MPPT control circuit is in the range of MHz.

This work proposes a low-complexity, closed-loop MPPT circuit for wide power range energy harvesting applications. It relies on a self-oscillating dc-dc boost (SOB) converter for the power conversion [21], which is controlled via an analog voltage. This allows the digital MPPT circuit to work with an extremely low clock frequency of 1 Hz, and consequently an ultra-low power consumption. The circuit works in a very wide input power range from 15 μ W up to 1 W with constant efficiency and is suitable for light energy harvesting both at indoor and outdoor lighting conditions.

II. PROPOSED MPPT CIRCUIT

A block diagram of the proposed MPPT circuit is shown in Fig. 1.

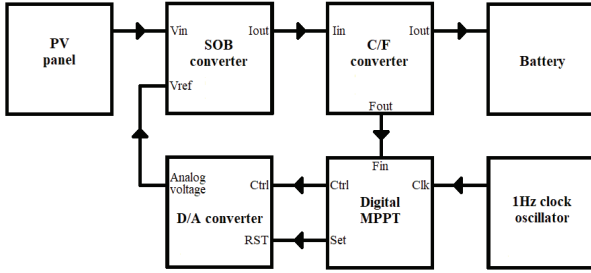


Figure 1. Block diagram of the proposed maximum power point tracking circuit

The circuit relies on a self-oscillating dc-dc boost (SOB) converter for the power conversion, described in a previous work [21]. The input impedance of the SOB converter can be controlled via an analog voltage. The digital part of the circuit uses a digital-to-analog (D/A) converter to produce the required analog voltage. Also, it uses feedback of the SOB converter's output power. As a large capacitance is used for the energy storage (rechargeable battery or supercapacitor), the output voltage level can be considered stable for short periods of time. Thus, the charging power is proportional to the charging current. This allows the circuit to measure the output power of the SOB converter using single sensor measurements of the output current. A resistor-free current-to-frequency (C/F) converter circuit is used to provide feedback of the SOB converter's output current.

A. Self-oscillating boost converter

The SOB converter (Fig. 2) is designed for efficient operation in a wide power range. The converter is self-oscillating and automatically adjusts its oscillating frequency to the input power.

The SOB converter provides input-impedance control, which is achieved using an external analog input pin (V_{ref}). The measured power conversion efficiency for the whole operating range (15 μ W – 1 W) and different MPP voltage levels (V_{MPP}) is shown in Fig. 3.

The control capability of the input impedance through an analog voltage is a very useful feature for MPPT implementation, as it reduces the complexity of the control circuit, compared with implementations that rely on width or frequency modulated pulses for the control (PWM/PFM). The SOB converter requires only 100 nA for its operation, as measured.

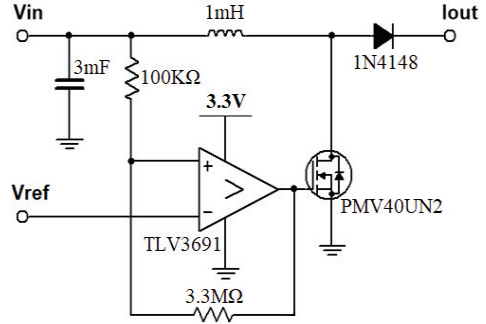


Figure 2. Schematic diagram of the self-oscillating dc-dc boost converter

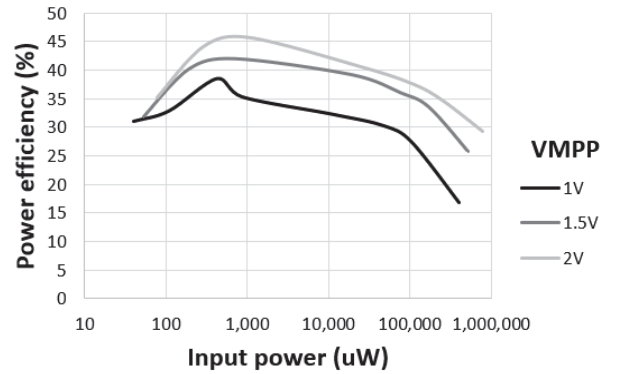


Figure 3. Power efficiency measurements of the self-oscillating dc-dc boost converter, versus the input power for different input voltage levels (V_{MPP})

B. Current to frequency converter

A C/F converter (Fig. 4) is used for the charging current sensing. The charging current enters through the I_{in} terminal and exits through the I_{out} terminal. A small fraction of the charging current is mirrored by the high side current mirror driving a current controlled op-amp based astable multivibrator. The circuit offers improved performance in comparison with resistor based current sensors, as it works with a constant voltage drop equal to $V(I_{in}) - V(I_{out}) = V_{BE} = 0.7$ V. Additionally, the C/F converter eliminates the need for amplification and analog to digital (A/D) conversion stages simplifying the overall design. The measured oscillating frequency of the C/F converter is shown in Fig. 5. As shown the converter

operates with good accuracy over a wide current range. Better performance can be achieved in case of on chip integration of the current mirror. The measured power consumption of the C/F converter is 185 nA for the voltage divider and 100 nA for the op-amp. Transistors with close gain values were selected for the current mirror and a low leakage Styrene capacitor was used for the astable multivibrator implementation.

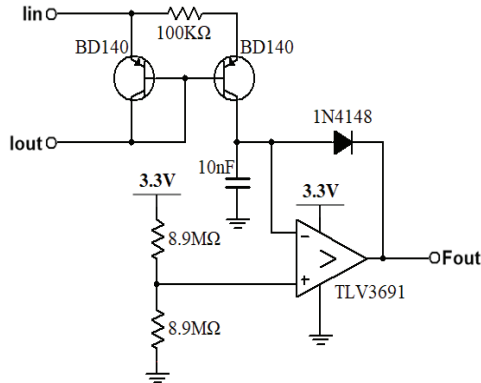


Figure 4. Schematic diagram of the current to frequency converter

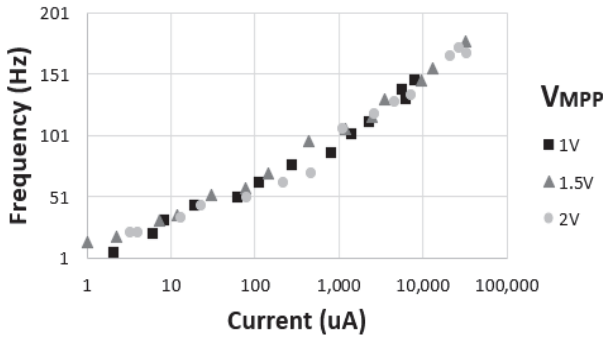


Figure 5. Frequency measurements of the current to frequency converter, versus the sensing current for different input voltage levels (V_{MPP})

C. Digital to analog converter

To control the input impedance of the SOB converter and perform MPPT, an analog voltage (V_{ref}) is required. A nano-power custom-made D/A converter was used to produce the *Analog voltage*, since the commercially available D/A ICs work with a higher power consumption. The design is built with the minimum components needed and uses a capacitor as the output element (Fig 6).

Using the *Ctrl* and *RST* digital input pins, the voltage of the capacitor can be controlled. When both signals are HIGH, the low side switch is active connecting the diode to the ground. The converter is in reset state and the output voltage is equal to the forward voltage of the diode ($V_F = 0.7$ V). When the *RST* signal is LOW, the charging or discharging state of the capacitor can be selected with the *Ctrl* signal.

Normally, feedback of the output *Analog voltage* is required for the converter to provide a stable output. Also,

the output should be buffered in order to be able to drive a load. In this design the digital controller is using feedback of the SOB converter's working power to control the charging or the discharging of the capacitor, and the *Analog voltage* level is oscillating around the MPP voltage. Finally, an output buffer is not required, as the D/A converter is used to drive the input of the op-amp of the SOB converter. The maximum current consumption of the D/A converter is 290 nA during the reset state.

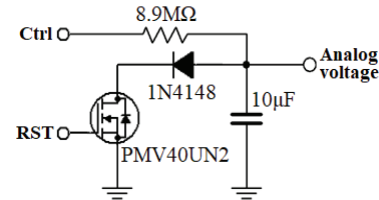


Figure 6. Schematic diagram of the digital to analog converter

D. Digital MPPT

The block diagram of the implemented digital MPPT module is shown in Fig 7.

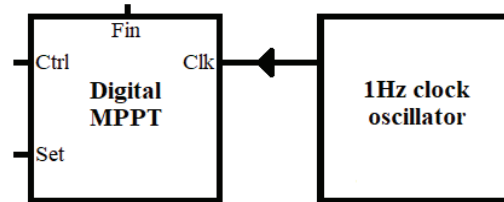


Figure 7. Block diagram of the FPGA digital control

The fact that the SOB converter does not require PWM or PFM control, allows the digital unit to operate with a very low clock frequency (1Hz), that minimizes the power consumption. In this work, the digital MPPT control has been described in Verilog HDL and implemented on an FPGA board. The FPGA also produces the 1Hz clock that is required and provides the power supply of the analog circuits (3.3V).

The digital control module is constantly measuring the frequency of the C/F converter, which is proportional to the output power of the SOB converter, using a digital input pin (*Fin*). Also, using two digital output pins (*Ctrl*, *Set*) it controls the D/A converter. The MPPT method used is the perturb and observe (P&O) logic. A P&O step is performed at each clock edge (*Clk* pin). The flow chart of the implemented code is shown in Fig.8.

The module uses two 12bit registers to store the current (*Cnt*) and previous (*LCnt*) counts of the C/F converter in order to decide the direction of the P&O step. Initially, the *Ctrl* and *Set* pins are set to HIGH and the analog voltage of the D/A converter is set at 0.7 V (reset state). As soon as energy is available from the PV panel, the SOB converter starts to oscillate charging the battery. The module monitors the pulses at the output of the C/F converter and initiates the P&O operation. The *Ctrl* signal

is set to HIGH, the *Set* signal is set to LOW, and the capacitor of the D/A converter is slowly charging. At each clock edge (*Clk*), the module is comparing the previous frequency counts (*LCnt*) with the current ones (*Cnt*). Assuming that the initial *Analog voltage* output of the D/A converter (0.7 V) is lower than the MPP voltage at this time, the measured counts should be increasing. When a decrement of the measured counts is detected, the module toggles the charging-discharging pin (*Ctrl*), and the circuit oscillates around the MPP voltage. When there is no available energy at the input, the oscillation of the SOB converter stops, and the module re-initializes and returns to the reset state loop.

The design consists of 47 logic gates and 29 flip-flop register elements.

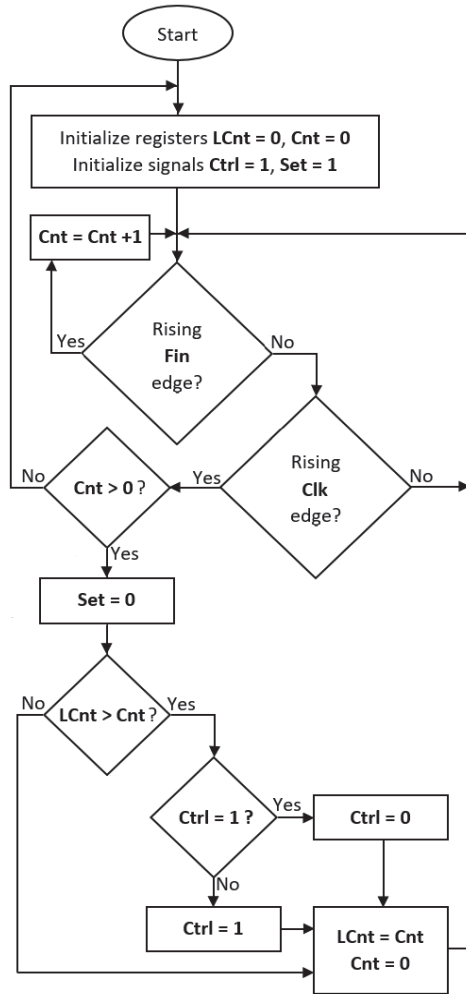


Figure 8. Flow chart of the code running on the FPGA

III. EXPERIMENTAL RESULTS

A. MPPT performance

To confirm the proper operation of the design, the proposed MPPT circuit was implemented and measured using discrete components for the analog part of the

circuit. The digital control module as well as the 1Hz clock oscillator have been described in Verilog HDL and implemented on an Intel Cyclone 10CL016 FPGA. Finally, the power supply of the analog circuits (3.3V) is provided by the FPGA board. The circuit implementation is shown in Fig. 9.

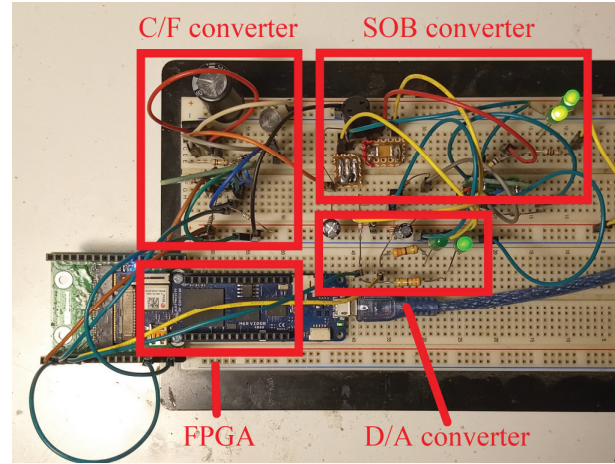


Figure 9. Discrete implementation of the proposed maximum power point tracking circuit

Fig. 10 shows the input voltage of the SOB converter (CH2) and the *Analog voltage* output of the D/A converter (CH1). For the first 15 seconds there is no energy available at the input of the SOB converter and the initial *Analog voltage* output is 0.7 V (reset state). Afterwards, energy is provided at the input of the SOB converter and the P&O operation starts.

The power conversion efficiency of the harvesting circuit was measured for the whole operating power range. Fig. 11 shows the power efficiency of the MPPT operation, and Fig. 12 shows the end-to-end power efficiency of the proposed MPPT circuit for different MPP voltages. As shown, the proposed circuit offers high MPPT efficiency and operates in a very wide power range from 15 uW up to 1 W. The design is suitable for PV panels with an MPP voltage ranging from 0.7 V up to 2.5 V (2 – 6 cells) and a maximum output power of 1 W. It can be used for light energy harvesting both at indoor and outdoor conditions.

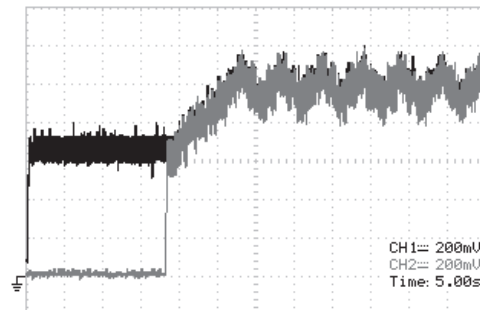


Figure 10. Oscilloscope view of the *Vref* analog voltage (CH1) and the input voltage (CH2) during the operation

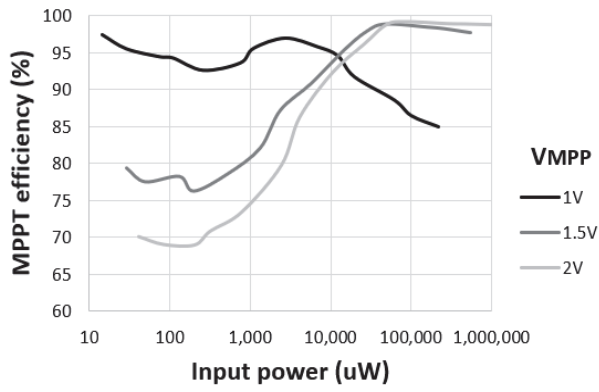


Figure 11. Maximum power point tracking efficiency versus the input power for different input voltage levels (V_{MPP})

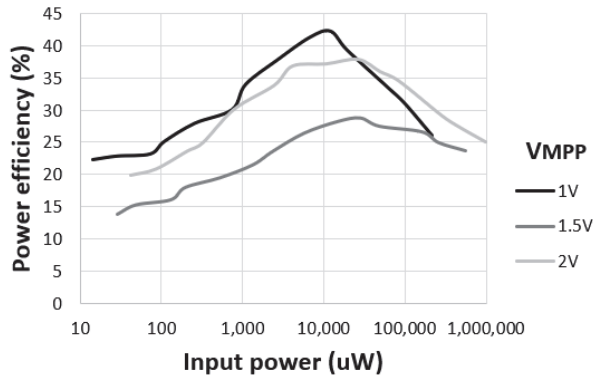


Figure 12. End-to-end efficiency versus the input power for different input voltage levels (V_{MPP})

B. Power consumption

The proposed MPPT circuit is optimized towards minimum power consumption. Even though this work is an early implementation of the design - for proofing reasons - power estimations seem promising in case of on chip integration. The overall current consumption, as well as the current consumption of each unit are gathered in Table I. The estimated power consumption for the proposed MPPT circuit is only 2.9 μ W at 3.3 V.

TABLE I. CURRENT CONSUMPTION OF EACH UNIT

Unit	Consumption
SOB converter	100 nA ¹
C/F converter	285 nA ¹
D/A converter	290 nA ¹
Digital MPPT	11 nA ²
1Hz clock	200 nA ³
Total	886 nA

¹ Measured; ² Estimated; ³ Simulated

The current consumptions of the analog units were measured using the discrete implementation of the circuit¹. The current consumption of the digital MPPT was estimated using modern standard-cell technology documentation files². An average static power dissipation of 0.5 nW/logic-element leads to an estimated current

consumption of 11 nA for the digital module [22]. An astable multivibrator oscillator based on an op-amp was designed and simulated in Cadence Virtuoso for the estimation of the 1Hz clock unit current consumption³.

C. Performance summary and comparison

In this section the proposed MPPT circuit is compared with other state-of-the-art designs. The working specifications and performance characteristics of each design are gathered in Table II and Table III, respectively.

TABLE II. MPPT DESIGN COMPARISON

Reference	Power sensing / Converter type	MPPT controller / Control type	Clock frequency
[5] ¹	Time-based / Boost	MCU / PFM	32.8 KHz
[6]	Time-based / Charge pump	Custom digital / Single shot	250 KHz
[7]	Voltage-current / Boost	Custom digital / PWM	1 MHz
[11]	Time-based / Charge pump	Custom digital / PFM	0.5 MHz
[14]	Voltage-current / Charge pump	Negative feedback analog VCO / PFM	10 MHz
[20]	Voltage-current / Buck	Custom digital / VCO PWM – PFM	0.5 MHz
This work	Time-based / SOB	Custom digital / Analog voltage	1 Hz

¹ Previous work

TABLE III. PERFORMANCE SUMMARY AND COMPARISON

Reference	Peak / Average MPPT Efficiency (%)	Power Throughput	Power Consumption
[5] ¹	99.9 / 99.75	10 μ W – 300 mW	73.5 μ W
[6]	86 / NA ²	0 – 21 μ W	12 μ W
[7]	99.9 / NA ²	0.4 – 21.1 W	11 mW
[11]	97 / 90	6 – 600 μ W	5.1 μ W
[14]	99.8 / NA ²	140 – 440 μ W	180 μ W
[20]	90 / 85	0.1 – 1 W	250 mW
This work	97 / 90	15 μW – 1 W	2.9 μW

¹ Previous work; ² Not available

As shown, the PWM-based designs, as well as the ones based on voltage-current power sensing, present the most limited working power range, presenting high peak efficiency at a specific input power value. On the other hand, designs based on PFM pulses and time-based power sensing achieve the widest working ranges with fair average efficiency. Additionally, as the MPPT circuit is also used to control the power converter, high clock frequencies are required for their operation.

The presented design is based on a self-oscillating converter and offers the widest operating range with high tracking accuracy. As the required clock is used to control only the MPPT flow, an extremely low frequency is selected, that drastically decreases the overall power consumption. Based on the proposed MPPT design, a

fully integrated indoor-outdoor light energy harvesting system can be implemented, with improved performance.

IV. CONCLUSIONS

A low-complexity FPGA-based light harvesting MPPT circuit with wide operating power range has been presented in this work. The design covers a very wide working power range from 15 μ W up to 1 W and is suitable for light energy harvesting both at indoor and outdoor conditions. The circuit was built and measured using discrete components and an FPGA board. Experimental results validate the proper operation of the proposed MPPT circuit with good accuracy and high tracking efficiency in the whole operating range. In case of on chip integration the proposed design can work with improved performance and very low power consumption.

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